

Formula-1: slide-3.1

Objects = n = Base HTML file(1) + Reference Objects(n-1)

Practice-1:

1. Phoenix has requested 10 objects. How many total objects are requested?
2. Phoenix has requested 10 objects including the base HTML file. How many total objects are requested?
3. Phoenix has requested 10 objects excluding the base HTML file. How many total objects are requested?
4. Phoenix has requested two jpeg images, three audio files excluding the base HTML file. How many total objects are requested?
5. Phoenix has requested two jpeg images, three audio files. How many objects are requested including the base HTML file?

Practice-2:

1. After fetching the IP address, device X has requested the web server to open a persistent HTTP connection with the server and request 28 objects.
How many TCP connections are required?
2. After fetching the IP address, device X has requested the web server to open a non-persistent HTTP connection with the server and request 30 objects.
How many TCP connections are required?
3. After fetching the IP address, device X has requested the web server to open a persistent HTTP connection with a pipeline size of 5 to the server and request 28 objects.
How many TCP connections are required?
4. After fetching the IP address, device X has requested the web server to open a non-persistent HTTP connection without pipelining to the server and request 30 objects.
How many TCP connections are required?

Formula-3: slide-3.1

- a. 1 RTT = Request+Response
- b. 1 RTT = 2 x One-way Transmission from A to B [If, Request=Response]
- c. Propagation Delay = One-way Transmission from A to B or vice-versa
- d. 1 HTTP RTT = HTTP request + HTTP response
- e. 1 TCP RTT = TCP connection request + TCP connection response or acknowledgement
- f. Ideal Case: 1 TCP RTT = 1 HTTP RTT (Web Request+Web Response)
- g. HTTP response = Database time + Server to Client response time

Practice-3:

1. After fetching the IP address, PC A sends the request to the website server (which takes 35ms to be sent).
What are the values of 1 TCP RTT and 1 HTTP RTT?
2. After fetching the IP address, PC A sends the request to the website server (which takes 25ms to be acknowledged only).
What are the values of 1 TCP RTT and 1 HTTP RTT?
3. After fetching the IP address, PC A requires 35 ms for TCP interactions to the website server and a HTTP RTT of 90 ms to open a HTTP connection.
What are the values of 1 TCP RTT and 1 HTTP RTT?
4. After fetching the IP address, PC A requires 35 ms for TCP interactions to the website server and 24 ms to make a HTTP request.
What are the values of 1 TCP RTT and 1 HTTP RTT?
5. After fetching the IP address, PC A requires a RTT of 35 ms for TCP interactions, 24 ms to make a HTTP request and 30 ms to get the HTTP response.
What are the values of 1 TCP RTT and 1 HTTP RTT?
6. After fetching the IP address, PC A sends the request to the website server (which takes 35ms to be sent) to open a HTTP connection. It takes the server 5ms to retrieve each object from the database and prepare it to be sent.
What are the values of 1 TCP RTT and 1 HTTP RTT?

Formula-4: slide-3.1

- a. Object fetch(after fetching the IP address) = TCP time + HTTP time
- b. Object fetch(before fetching the IP address or from the point of requesting the website/ entering the URL) = DNS time + TCP time + HTTP time

Practice-4:

1. After fetching the IP address, Phonte's PC sends the request to the website server (which takes 21ms to be sent only) to open a persistent HTTP connection with the server and request 39 objects, including the base HTML page. (DNS time = X ms)
Calculate the total RTT required to fetch all the objects after retrieving the IP address.
2. After fetching the IP address, Saif's PC sends the request to the website server (which takes 39ms to be sent) to open a non-persistent HTTP connection with the server and request 30 objects, including the base HTML page. It takes the server 4ms to retrieve each object from the database and prepare it to be sent. (DNS time = X ms)
Calculate the total RTT required to fetch all the objects from the point of requesting the website.

Formula-5: slide-3.1

- a. Webpage load time or total response time = DNS + TCP + HTTP + TT
- b. Webpage load time(after fetching the IP address) = TCP + HTTP + TT

Practice-5:

1. After fetching the IP address, Saif's PC sends the request to the website server (which takes 39ms to be sent) to open a non-persistent HTTP connection with the server and request 30 objects, including the base HTML page. It takes the server 4ms to retrieve each object from the database and prepare it to be sent. Each object takes 215ms to be downloaded by the client. (DNS time = X ms)

Calculate the total time PC A takes to load the webpage.

2. After fetching the IP address, PC B sends the request to the website server (which takes 25ms to be sent) to open a persistent HTTP connection with the server and request 12 objects, including the base HTML page, each requiring 85ms to be downloaded. (DNS time = X ms)

Calculate the total time PC B takes to load the webpage.

persistent without pipelining HTTP connection to the server(taking 20ms to be sent only). If she was requesting for 15 objects, then how much time will she have to wait from her entering the URL of Valorant in the browser till loading the whole website on her PC for her second time access request(IP address not expired), if first 10 objects take 244ms to be downloaded and the rest of the objects take 56ms to be downloaded? (DNS time = X ms)